# 11. Equations with Prescribed Group

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The problem of constructing equations with prescribed group can be attacked in two directions, which one may briefly call the "irrational" direction, characterizing the roots, and the "rational" direction, characterizing the coefficients.

In the "irrational" direction, which works function-theoretically and arithmetically, lies Kronecker's theorem that all Abelian fields in the domain of the rational numbers are cyclotomic fields, and the corresponding theorems for relatively Abelian fields with respect to quadratic number fields. These theorems give, on the one hand, the realizability of all Abelian groups with respect to these special domains of rationality; on the other hand, they supply the totality of the equations and at the same time give a deeper insight into the structure of the number fields thereby defined. Each individual theorem, however, is restricted to a special domain of rationality; and every new domain of rationality requires an entirely new treatment.

The "rational", algebraic direction rests on Hilbert's irreducibility theorem, according to which in the coefficient representation one may restrict oneself to a parameter representation. To this direction belongs, for example, the existence of arbitrarily many equations with alternating group with respect to any prescribed domain of rationality. Here the insight, described above, into the structure of the fields defined by the equations is naturally absent; but the advantage of this "rational" direction lies in the fact that the realizability of the groups is proved for *all domains of rationality simultaneously*; nevertheless, the parameter representations known up to now do not in general yield the totality of the equations.[1]

What follows is a contribution to the "rational" direction; *sufficient conditions* are given for the *existence of parameter representations which, for any domain of rationality, supply the totality of the equations with prescribed group.* These conditions consist in this: the invariant field belonging to the group – Lagrange's genus domain – can be represented rationally by independent functions of the domain, that is, it possesses a "minimal basis"; or, expressed otherwise, it is isomorphic to the field of all rational functions of $n$ indeterminates. As will still be shown in § 2, this is satisfied, for example, for all groups occurring in equations of the third and fourth degrees. The actual construction and more exact discussion of these equations of the third and fourth degrees is found in the dissertation of F. Seidelmann,[2] an extract from which immediately follows this communication.

The question of the minimal basis of Lagrange's genus domains was – as already mentioned in the paper "Körper und Systeme rationaler Funktionen" (Math. Ann. 76) – raised to me by E. Fischer independently of the connection just described, and gave the impetus for the present investigations.[3]

## § 1.

### Sufficient Conditions for the Parameter Representation.

1. The possibility of reducing the construction of equations with prescribed group to parameter representation follows from Hilbert's irreducibility theorem, which says the following. Suppose that the equation

$$f(x) = x^n + F_1(\lambda_1 \cdots \lambda_r)x^{n-1} + \cdots + F_n(\lambda_1 \cdots \lambda_r) = 0 \tag{1}$$

– where $F_i(\lambda_1 \cdots \lambda_r)$ denotes rational functions of the independent parameters $\lambda_1 \cdots \lambda_r$, with coefficients from a number field[4] $\Omega$ – has, with respect to the field $\Omega(\lambda_1 \cdots \lambda_r)$ of all rational functions of $\lambda_1 \cdots \lambda_r$ with coefficients from $\Omega$, the group $\Gamma$. Then one can replace the parameters $\lambda$ by arbitrarily many rational numbers in such a way that the resulting numerical equation

$$g(x) = x^n + a_1x^{n-1} + \cdots + a_n = 0 \tag{2}$$

[1] For solvable equations, the parameter representation of the coefficients can be replaced by that of the roots. Recently F. Mertens has given, for certain groups of degree 8, most general root representations: "Gleichungen 8ten Grades mit Quaternionengruppen", Sitzb. d. Ak. d. Wiss., Wien, Abt. IIa, Bd. 125, p. 735. As I gather from that note, G. Bucht had already earlier given the most general root expressions for always constructible normal forms of equations of degree 3 and 4 and of equations of degree 8 with the groups mentioned: "Über einige algebraische Körper achten Grades", Arkiv for Math., Astron. och Physik, Bd. 6, Nr. 30. [Addition at correction.]

[2] Die Gesamtheit der kubischen und biquadratischen Gleichungen mit Affekt bei beliebigem Rationalitätsbereich. Erlangen 1916.

[3] Cf. Part 2 of the preliminary communication on "Rationale Funktionenkörper", Jahresb. d. d. Math. Vereinig., Bd. 22, 1913.

[4] Instead of the number field, a general domain of rationality could also occur.

has, with respect to $\Omega$, precisely the group $\Gamma$. Such a $g(x)$ shall be denoted more precisely by $g_\Gamma$.

The question now is whether *such a parameter representation* (1) *can be given that the resulting numerical equation* (2) *runs through the totality of all* $g_\Gamma$ *when* $\lambda_1 \cdots \lambda_r$ *run through all quantities from* $\Omega$ – with the exclusion of those quantities from $\Omega$ which cause a reduction of the group.[5] In other words: the nonlinear system of equations

$$F_1(\lambda_1 \cdots \lambda_r) = a_1; \quad \cdots ; \quad F_n(\lambda_1 \cdots \lambda_r) = a_n \tag{3}$$

should possess, for every system of values $a_1 \cdots a_n$ corresponding to a $g_\Gamma$, a solution, and indeed in quantities $\lambda_1 \cdots \lambda_r$ from $\Omega$. Since the system (3) is nonlinear, one must from the outset introduce a restriction when demanding that the totality of the $g_\Gamma$ be obtained through a parameter representation. Namely, in such nonlinear systems there can always occur singular systems of values $a_1 \cdots a_n$, satisfying an algebraic relation $H(a_1 \cdots a_n) = 0$, for which *no* solution exists;[6] for them the parameter representation (1) fails and a supplementary representation is necessary. But in the present case these singular values of $a_1 \cdots a_n$, as will be shown, can be characterized simply.

2. To obtain such a parameter representation, we impose the stronger requirement that the $\lambda_i$ be *natural irrationalities belonging to the group, identically in the roots regarded as indeterminates.* By this we mean the following. If in (1) one replaces the parameters $\lambda_1 \cdots \lambda_r$ by suitably chosen *rational* functions $\varphi_1(x) \cdots \varphi_r(x)$ of the indeterminates $x_1 \cdots x_n$ – with coefficients from $\Omega$ – then (1) is to pass into

$$h(x) = x^n - \sigma_1(x)x^{n-1} + \sigma_2(x)x^{n-2} \cdots \pm \sigma_n(x), \tag{4}$$

where $\sigma_1(x) \cdots \sigma_n(x)$ denote the elementary symmetric functions of $x_1 \cdots x_n$; and $h(x)$, with respect to the domain of rationality $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ thereby arising from $\Omega(\lambda_1 \cdots \lambda_r)$, is to have the group $\Gamma$. Because of the independence of the parameters, $\varphi_1(x) \cdots \varphi_r(x)$ are also to be algebraically independent functions.

To make this requirement precise, one must clarify the connection of $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ with the group $\Gamma$. For this we first determine the most general domain of rationality $K$ with respect to which $h(x)$ becomes $h_\Gamma$. Let $\mathfrak{L}_\Gamma$ denote the field of all rational functions with rational integral coefficients of $x_1 \cdots x_n$ which admit $\Gamma$ – also called the "Lagrangian genus domain" belonging to $\Gamma$, or the invariant field of $\Gamma$ – and let $\Omega(\mathfrak{L}_\Gamma)$ denote the corresponding field arising by adjoining $\mathfrak{L}_\Gamma$ to $\Omega$. Thus $\mathfrak{L}_\Gamma$ is a subfield of the field $\mathfrak{R}(x_1 \cdots x_n)$ of all rational functions with rational integral coefficients of $x_1 \cdots x_n$. According to the definition of the group, the most general domain $K$ is then given by *any field which contains* $\mathfrak{L}_\Gamma$ *as a subfield and whose intersection with* $\mathfrak{R}(x_1 \cdots x_n)$ *is exactly* $\mathfrak{L}_\Gamma$. Correspondingly, for every domain $K$ containing $\Omega$, the intersection with $\Omega(x_1 \cdots x_n)$ is given by $\Omega(\mathfrak{L}_\Gamma)$. Therefore, in particular, since by our requirement $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is to be a domain $K$, the intersection of $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ with $\Omega(x_1 \cdots x_n)$ is given by $\Omega(\mathfrak{L}_\Gamma)$; and since on the other hand $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is a subfield of $\Omega(x_1 \cdots x_n)$, $\Omega(\varphi_1(x) \cdots \varphi_r(x))$ is exactly equal to $\Omega(\mathfrak{L}_\Gamma)$. The requirement therefore says that *Lagrange's genus domain* $\Omega(\mathfrak{L}_\Gamma)$ *is to arise by adjoining the algebraically independent functions* $\varphi_1(x) \cdots \varphi_r(x)$ *to* $\Omega$; here one must have $r = n$, since $\mathfrak{L}_\Gamma$ contains the $n$ algebraically independent elementary functions, while on the other hand no more than $n$ functions can be algebraically independent. The functions $\varphi_1(x) \cdots \varphi_n(x)$ shall be said, as we shall also put it, to form a *minimal basis* of $\Omega(\mathfrak{L}_\Gamma)$; or also, $\mathfrak{L}_\Gamma$ possesses a minimal basis with coefficients from $\Omega$.

3. It is now essential that this whole consideration can be reversed, and thus actually leads to sufficient conditions[7] for the desired parameter representation. We therefore assume that $\varphi_1(x) \cdots \varphi_n(x)$ is a minimal basis of $\Omega(\mathfrak{L}_\Gamma)$; and let $\Omega$ be the smallest field containing the coefficients of $\varphi_1(x) \cdots \varphi_n(x)$. Then $\Omega$ is necessarily an algebraic number field, and can in particular be the field $\mathfrak{R}$ of all rational numbers. Since $\sigma_1(x), \ldots, \sigma_n(x)$ belong to $\Omega(\mathfrak{L}_\Gamma)$, there exist identities in $x_1 \cdots x_n$ – where the $F_i$ denote rational functions of their arguments:

$$\sigma_1(x) = F_1(\varphi_1(x) \cdots \varphi_n(x)); \quad \cdots ; \quad \sigma_n(x) = F_n(\varphi_1(x) \cdots \varphi_n(x)); \tag{5}$$

whereas conversely $\varphi_1(x) \cdots \varphi_n(x)$, and hence also $\varphi(x) = \mu_1\varphi_1(x) + \cdots + \mu_n\varphi_n(x)$, depend algebraically on $\sigma_1(x) \cdots \sigma_n(x)$ by means of the equation irreducible with respect to $\Omega(\sigma_1(x) \cdots \sigma_n(x))$:

$$G(\varphi(x); \sigma_1(x) \cdots \sigma_n(x)) = 0, \tag{6}$$

which is again an identity in the $x$.

[5]The occurrence of a double root of $g(x) = 0$ is also to be included here.

[6]This is to be carried out shortly in general in a paper on the "alternative in nonlinear systems of equations."

[7]The requirement in 2. was enlarged relative to 1., so that the conditions need not be necessary.

By virtue of (5), this identity (6) passes into

$$G(\varphi(x); F_1(\varphi_1 \cdots \varphi_n); \cdots ; F_n(\varphi_1 \cdots \varphi_n)) = H(\varphi_1(x) \cdots \varphi_n(x)) = 0; \tag{7}$$

and because of the algebraic independence of $\varphi_1(x) \cdots \varphi_n(x)$, (7) is fulfilled not only identically in $x$, but also in the $\varphi_i$; that is, identically in independent parameters $\lambda$ one has

$$H(\lambda_1 \cdots \lambda_n) = G(\mu_1\lambda_1 + \cdots + \mu_n\lambda_n; F_1(\lambda_1 \cdots \lambda_n); \cdots ; F_n(\lambda_1 \cdots \lambda_n)) = 0. \tag{8}$$

From the identity (8) it follows that the equation

$$f(x) = x^n - F_1(\lambda_1 \cdots \lambda_n)x^{n-1} + F_2(\lambda_1 \cdots \lambda_n)x^{n-2} \cdots \pm F_n(\lambda_1 \cdots \lambda_n) = 0 \tag{9}$$

*has the group* $\Gamma$ *with respect to* $\Omega(\lambda_1 \cdots \lambda_n)$. For this identity shows that the rationally known $\lambda = \mu_1\lambda_1 + \cdots + \mu_n\lambda_n$ is a function of the roots belonging to $\Gamma$; but no further function belonging to a subgroup of $\Gamma$ can be rationally known, as is shown by the substitution $\lambda_i = \varphi_i(x)$; and under the same substitution $\lambda$ is also different from all its conjugates. If further $\Omega^*$ is any number field containing $\Omega$, then $f(x)$ *has the group* $\Gamma$ *with respect to* $\Omega^*(\lambda_1 \cdots \lambda_n)$; for by 2. the group is not reduced under the substitution $\lambda_i = \varphi_i(x)$ even after adjoining $\Omega^*$.

It remains to prove that, in the sense modified in 1., $f(x)$ actually runs through the totality of all $g_\Gamma$. For this, observe that by (5) the $F_i(\lambda)$ are algebraically independent functions of their arguments $\lambda$. The system of equations

$$F_1(\lambda_1 \cdots \lambda_n) = -a_1; \quad F_2(\lambda_1 \cdots \lambda_n) = a_2; \quad \cdots ; \quad F_n(\lambda_1 \cdots \lambda_n) = \pm a_n \tag{10}$$

therefore possesses solutions for every arbitrary system of values $a_1 \cdots a_n$ – with the exception of the singular ones still to be specified. For every system of values $a_1 \cdots a_n$ corresponding to a $g_\Gamma$ with respect to $\Omega^*$, the corresponding $\lambda_i$, as natural irrationalities belonging to the group,[8] become quantities from $\Omega^*$. Thus as $\lambda_1 \cdots \lambda_n$ run through all systems of values, $g(x)$ runs through the totality of all equations (in the modified sense); and this totality contains the totality of the $g_\Gamma$ to which values from $\Omega^*$ correspond. Since, conversely, by Hilbert's irreducibility theorem the values $\lambda$ from $\Omega^*$ "in general" also correspond to equations $g_\Gamma$, one actually obtains, in the sense modified in 1., through the parameter representation (9) the totality of all $g_\Gamma$ with respect to $\Omega^*$.

It remains now to determine the singular systems of values for which the parameter representation fails. Split into numerator and denominator, let the functions of the minimal basis be given by

$$\varphi_i(x) = \frac{\psi_i(x)}{\chi_i(x)}.$$

Substituting this into the identities (5), suppose that these identities become, without cancellation,

$$\sigma_k(x) = \frac{G_k(x)}{H_k(x)} \quad \text{or} \quad H_k(x) \cdot \sigma_k(x) = G_k(x), \tag{11}$$

so that the product over all $H_k(x)$ is divisible by the product of all denominators $\chi_i(x)$. For all root systems $\alpha_1 \cdots \alpha_n$ for which the product

$$H_1(\alpha) \cdot H_2(\alpha) \cdots H_n(\alpha) \neq 0$$

and consequently no denominator $\chi_i(\alpha)$ vanishes either, one can divide (11) by $H_k(\alpha)$ and obtains

$$\sigma_k(\alpha) = \frac{G_k(\alpha)}{H_k(\alpha)} = F_k(\varphi_1(\alpha); \cdots ; \varphi_n(\alpha)), \tag{12}$$

where the $\varphi_i(\alpha)$, because of the nonvanishing of the denominators, assume finite values – and for equations $g_\Gamma$ become quantities from $\Omega^*$. Equation (12) represents the solution system of (10), as soon as $\alpha_1 \cdots \alpha_n$ are determined so that $a_k = (-1)^k\sigma_k(\alpha)$.[9]

[8] Cf. also below!

[9] The solution of the equation is thereby reduced to the solution of the system, consisting of independent functions,

$$\varphi_1(\alpha_1 \cdots \alpha_n) = \lambda_1; \quad \cdots ; \quad \varphi_n(\alpha_1 \cdots \alpha_n) = \lambda_n.$$

Thus only such coefficient systems $a_k = (-1)^k \sigma_k(\alpha)$ can become *singular* for which

$$H_1(\alpha) \cdot H_2(\alpha) \cdots H_n(\alpha) = 0;$$

for these, (11), instead of becoming a representation, passes into $0 = 0$. A special investigation is then needed as to whether among the singular $a_1 \cdots a_n$ so defined there are also such which correspond to equations $g_\Gamma$, and for which number fields.[10] In summary, we obtain the following.

*Theorem. If the Lagrangian genus domain belonging to the group $\Gamma$ possesses a minimal basis*[11] *with coefficients from $\Omega$, then for every number field $\Omega^*$ containing $\Omega$, the totality of equations with prescribed group $\Gamma$ with respect to $\Omega^*$ can be constructed rationally by the parameter representation (2) – possibly with the exception of those which are singular with respect to the parameter representation and can be given separately. The parameters are to run through all values from $\Omega^*$ which do not cause a reduction of the group. The construction is possible for every number field if the coefficient field $\Omega$ of the minimal basis is equal to the field $\mathfrak{R}$ of all rational numbers.*

Since, by Hilbert's irreducibility theorem, the parameter manifold is not lowered by excluding those values from $\Omega^*$ which cause a reduction of the group, it follows further: *from the existence of a minimal basis it follows that the manifold of equations with the group $\Gamma$ with respect to $\Omega^*$ is equal to the manifold of the affect-free equations; the manifold is not lowered by the affect.*

## § 2.

## Application to Special Equations.

We now prove the existence of the minimal basis for all groups occurring in equations of the third and fourth degrees; and for this we first show quite generally that, for equations of degree $n$, the problem can be reduced to one of $n-2$ indeterminates. The first reduction corresponds to the linear Tschirnhaus transformation for removing the second-highest term. We take as first basis function $\varphi_1(x) = \sigma_1(x)$; and we observe further that $\mathfrak{L}_\Gamma$ can be obtained rationally from all *homogeneous* forms in $x_1 \cdots x_n$ admitting $\Gamma$. Let $p(x_1 \cdots x_n)$ be such a form; then also

$$q(x) = p\left(x_1 - \frac{\sigma_1(x)}{n}; \cdots; x_n - \frac{\sigma_1(x)}{n}\right) = p\left(x_i - \frac{\sigma_1(x)}{n}\right) = p(y_i)$$

belongs to $\mathfrak{L}_\Gamma$, since it admits $\Gamma$, and moreover one has

$$p(x) \equiv p\left(x_i - \frac{\sigma_1(x)}{n}\right) \bmod \sigma_1(x)$$

or

$$p(x) = q(x) + \sigma_1(x) \cdot p_1(x)$$

with $p_1(x)$ belonging to $\mathfrak{L}_\Gamma$, and of smaller dimension than $p(x)$. Continuing the procedure thus shows that all homogeneous forms from $\mathfrak{L}_\Gamma$ – and consequently all rational functions from $\mathfrak{L}_\Gamma$ at all – can be rationally expressed by $\varphi_1(x) = \sigma_1(x)$ and by homogeneous forms

$$q(x) = p\left(x_i - \frac{\sigma_1(x)}{n}\right) = p(y_i).$$

Between these quantities $y_i$ there is, however, the linear relation $\sum y_i = 0$; hence the $q(x)$ *depend only on* $(n-1)$ *arguments.*

The second reduction rests on the fact that the $q(x)$ are homogeneous forms of their arguments. We take as second basis function

$$\varphi_2(x) = \frac{\tau_3(x)}{\tau_2(x)}, \quad \text{where} \quad \tau_3(x) = \sigma_3\left(x_i - \frac{\sigma_1(x)}{n}\right) = \sigma_3(y_i), \quad \tau_2(x) = \sigma_2\left(x_i - \frac{\sigma_1(x)}{n}\right) = \sigma_2(y_i);$$

[10] For example, as F. Seidelmann has shown at the cited place, only for the alternating group of equations of the third and fourth degrees do such equations – corresponding to singular systems of values – occur, and indeed only for those number fields $\Omega^*$ which contain the field of third roots of unity. In each case a simple supplementary representation can be given.

[11] From the existence of one minimal basis there follows at once the existence of arbitrarily many, derivable from the original one by reversible rational transformations.

thus $\varphi_2(x)$ is a homogeneous fractional function of first degree of $(n-1)$ arguments, namely of the $y_i$ with $\sum y_i = 0$. But by means of $\varphi_2(x)$ all $q(x)$ can be expressed rationally by the functions, likewise belonging to $\mathfrak{L}_\Gamma$,

$$r(x) = \frac{q(x)}{\varphi_2(x)^\lambda} = \frac{q(x)\cdot\tau_2(x)^\lambda}{\tau_3(x)^\lambda} = \frac{p(y_i)\cdot\sigma_2(y_i)^\lambda}{\sigma_3(y_i)^\lambda},$$

where $\lambda$ denotes the degree of $q(x)$. Thus $r(x)$ is homogeneous of degree zero and therefore depends only on $(n-2)$ arguments, the ratios $y_1 : y_2 : \cdots : y_n$ with $\sum y_i = 0$. Hence $\mathfrak{L}_\Gamma$ can be rationally expressed by

$$\varphi_1(x) = \sigma_1(x); \quad \varphi_2(x) = \frac{\tau_3(x)}{\tau_2(x)}$$

and by the *field of all functions $r(x)$ from $\mathfrak{L}_\Gamma$ depending on these $(n-2)$ arguments*,[12] whereby the desired reduction is accomplished.

For equations of the third and fourth degrees, this gives in particular a reduction to function fields of one and two arguments respectively. But for these a minimal basis always exists by the theorems of Lüroth and Castelnuovo.[13] In fact, in the case of one indeterminate the minimal basis can be derived by rational operations, and therefore, specifically for $\mathfrak{L}_\Gamma$, contains only rational numerical coefficients. In the case of two indeterminates, according to Castelnuovo, one might still be dealing with a basis with coefficients from an algebraic number field $\Omega$; the actual investigation (cf. F. Seidelmann, loc. cit.) shows that here too, for every $\mathfrak{L}_\Gamma$, one obtains a basis with rational integral coefficients. One sees this almost without computation already from the fact that for the four group such a rational-integral basis can be chosen, namely

$$\varphi_1(x) = \sigma_1(x); \quad \varphi_2(x) = \frac{vw}{u}; \quad \varphi_3(x) = \frac{u\cdot w}{v}; \quad \varphi_4(x) = \frac{u\cdot v}{w},$$

where

$$u = x_1 + x_2 - x_3 - x_4; \quad v = x_1 - x_2 + x_3 - x_4; \quad w = x_1 - x_2 - x_3 + x_4,$$

such that the alternating group passes into the cyclic group of the three basis functions $\varphi_2, \varphi_3, \varphi_4$, and each group of order eight into the symmetric group of two of these functions, whereby a reduction to the groups of equations of the third and second degrees is effected. For the cyclic group, however, a rational-integral basis is already known from Abel, although not formulated as such.[14] Thus it is proved: *The totality of equations of the third and fourth degrees with prescribed group – possibly with the exception of those singular with respect to the parameter representation – can be constructed rationally by parameter representation for any prescribed domain of rationality; their manifold is equal to the manifold of affect-free equations.*

The actual investigation shows (cf. F. Seidelmann, loc. cit.) that a supplementary representation is needed only for the alternating group and only with respect to domains of rationality containing the field of third roots of unity, whereas in all other cases the parameter representation supplies the totality without qualification.

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[12] For an equation of degree $n$ without second term,

$$f(x) = x^n + a_2x^{n-2} + a_3x^{n-3} + \cdots + a_n = 0,$$

this second reduction corresponds to the substitution, always possible for $a_2 \neq 0$, $a_3 \neq 0$,

$$y = \frac{x\cdot a_2}{a_3},$$

by which $f(x)$, up to a factor, passes into

$$g(y) = y^n + \frac{a_2^3}{a_3^2}y^{n-2} + \frac{a_2^3}{a_3^2}y^{n-3} + \frac{a_4\cdot a_2^4}{a_3^4}y^{n-4} + \cdots + a_n\frac{a_2^n}{a_3^n} = 0;$$

that is, into an equation containing only $n-2$ parameters:

$$g(y) = y^n + c\cdot y^{n-2} + c\cdot y^{n-3} + c_4y^{n-4} + \cdots + c_n = 0.$$

[13] J. Lüroth: Beweis eines Satzes über rationale Kurven, Math. Ann. 9 (1875); G. Castelnuovo: Sulla razionalità delle involuzioni piane, Math. Ann. 44 (1893). That for fields of more than two indeterminates a minimal basis need not exist was shown by F. Enriques by a counterexample, Rend. Acc. Linc., Vol. 21, 21 Jan. 1912.

[14] Cf. Weber Algebra (small edition), § 93, Formula (12).

It should also be noted that the minimal basis is known for all *Abelian groups.*[15] Here, however, the basis has coefficients from the cyclotomic field derived from the group characters. Thus, in this case, no new results can be obtained for the construction of equations with prescribed group.

Göttingen, July 1916.

[15] E. Fischer: Die Isomorphie der Invariantenkörper der endlichen Abelschen Gruppen linearer Transformationen. Gött. Nachr. 1915, and Zur Theorie der endlichen Abelschen Gruppen. Math. Ann. 77 (1915).